

Slipping rope — Solution

W24 (2024) — English translation

A1

1.50 pts

Determine the speed v and acceleration a of the left end of the rope at the moment when the left end is lowered by the amount x . Express your answer using g , L_1 , L , α and x .

First solution: The length of the part of the rope hanging on the right is $L_2 = L_1 + L \sin \alpha$, and the total length of the rope is:

$$l = 2L_1 + L(1 + \sin \alpha).$$

Using the law conservation mechanical energy:

$$\frac{mv^2}{2} = -\Delta W_p,$$

where $m = \lambda l$ - mass rope, and/but ΔW_p - change of potential energy in field тяжести. Change potential energy rope in field тяжести equal change potential energy section rope length x , coordinate x_C of the center of mass which изменилась from value $x_{C_1} = -x/2$ to $x_{C_2} = x/2$. Hence have:

$$\Delta W_p = -\lambda x \cdot g \cdot (x_{C_2} - x_{C_1}) = -\lambda g x^2.$$

Substitute ΔW_p in conservation of mechanical energy, obtain:

$$\frac{mv^2}{2} = \frac{\lambda l v^2}{2} = \lambda g x^2 \Rightarrow v = x \sqrt{\frac{2g}{l}},$$

from:

$$l = 2L_1 + L(1 + \sin \alpha).$$

Let's use the law of conservation of mechanical energy:

$$\frac{mv^2}{2} = -\Delta W_p,$$

where $m = \lambda l$ is the mass of the rope, and ΔW_p is the change in its potential energy in the gravitational field.

The change in the potential energy of the rope in the gravitational field is equal to the change in the potential energy of a section of the rope of length x , the coordinate x_C of the center of mass of which has changed from the value $x_{C_1} = -x/2$ to $x_{C_2} = x/2$. From here we have:

$$\Delta W_p = -\lambda x \cdot g \cdot (x_{C_2} - x_{C_1}) = -\lambda g x^2.$$

Substituting ΔW_p into the law of conservation of mechanical energy, we obtain:

$$\frac{mv^2}{2} = \frac{\lambda l v^2}{2} = \lambda g x^2 \Rightarrow v = x \sqrt{\frac{2g}{l}},$$

where:

Since the speed $v(x)$ and the displacement x are related by direct proportionality:

$$a = v \sqrt{\frac{2g}{l}} = \frac{2gx}{l},$$

from where:

Second solution: Let's write down the equation of motion for each section of the rope:

$$\lambda(L_1 + x)a = \lambda(L_1 + x)g - T_1 \quad \lambda La = T_1 - T_2 + \lambda g L \sin \alpha \quad \lambda(L_2 - x)a = T_2 - \lambda(L_2 - x)g,$$

where T_1 and T_2 - force tension rope near block 1 and 2 correspondingly. Add equation, obtain:

$$\lambda(L_1 + L_2 + L)a = \lambda g(L_1 + L \sin \alpha - L_2 + 2x),$$

from where

$$\lambda(L_1 + x)a = \lambda(L_1 + x)g - T_1 \quad \lambda La = T_1 - T_2 + \lambda g L \sin \alpha \quad \lambda(L_2 - x)a = T_2 - \lambda(L_2 - x)g,$$

where T_1 and T_2 are the rope tension forces near blocks 1 and 2, respectively.

Adding the equations, we get:

$$\lambda(L_1 + L_2 + L)a = \lambda g(L_1 + L \sin \alpha - L_2 + 2x),$$

where

Multiplying the expression for acceleration by speed, we get:

$$av = \frac{d(v^2/2)}{dt} = \frac{2gxv}{2L_1 + L(1 + \sin \alpha)} = \frac{g}{2L_1 + L(1 + \sin \alpha)} \frac{d(x^2)}{dt}$$

Integrate, obtain:

$$\frac{v^2}{2} = \frac{gx^2}{2L_1 + L(1 + \sin \alpha)},$$

from:

Answer: Answer:

$$v = x \sqrt{\frac{2g}{2L_1 + L(1 + \sin \alpha)}}.$$

A2

0.80 pts

Determine the amount of displacement of the left end of the rope x_1 at the moment in time when the tension of the section of the rope lying on the inclined plane is constant along the length of this section. Express your answer in terms of L_1 , L and α . At what ratios L_1 , L and α is the value x_1 less than the initial length of the right section of the rope?

Let us introduce the axis y , directed along the inclined plane from block 1 to block 2. Let us write in projection onto the axis y Newton's second law for a section of rope of length dy lying on an inclined plane:

$$dT + \lambda g \sin \alpha dy = \lambda a dy.$$

Quantity $dT = 0$ for $a(x_1) = g \sin \alpha$. Hence:

$$\frac{2gx_1}{l} = g \sin \alpha \Rightarrow x_1 = \frac{l \sin \alpha}{2},$$

from where:

Let's compare the value x_1 with the value L_2 :

$$x_1 = L_1 \sin \alpha + \frac{L \sin \alpha (1 + \sin \alpha)}{2} \quad L_2 = L_1 + L \sin \alpha.$$

Coefficient before L_1 and L in expression for L_2 larger corresponding coefficient in expression for x_1 , therefore:

Answer: Answer:

$$x_1 = L_1 \sin \alpha + \frac{L \sin \alpha (1 + \sin \alpha)}{2}.$$

A3

1.50 pts

Determine the maximum tension force of the rope at the moment of time when its left end has shifted by a distance x . Express your answer in terms of λ , g , L_1 , L , α and x .

On vertical sections of the rope, the greatest tension force is achieved at the points of contact of the blocks, since the acceleration $a(x)$ is always less than g . Let us denote by T_1 and T_2 the tension force of the rope at the points of contact of blocks 1 and 2, respectively. From Newton's second law for the left vertical section of the rope we obtain:

$$\lambda(L_1 + x)g - T_1 = \lambda(L_1 + x)a \quad T_2 - \lambda(L_2 - x)g = \lambda(L_2 - x)a,$$

from which:

$$T_1 = \lambda(L_1 + x)(g - a) \quad T_2 = \lambda(L_2 - x)(g + a).$$

Проанализируем inclined section rope. From solution part A2 follow:

$$\frac{dT}{dy} = \lambda(a - g \sin \alpha).$$

For $x \leq x_1$ quantity dT/dy not positive, and/but for $x \geq x_1$ - неотрицательна, from which follow, that:

$$T_{max} = \begin{cases} T_1 & \text{for } x \geq x_1 \\ T_2 & \text{for } x \leq x_1 \end{cases}$$

Substitute expression for a , L_2 and x_1 , we get:

Answer: Answer:

$$T_{max} = \begin{cases} \frac{\lambda g(L_1 + x)(2L_1 + L(1 + \sin \alpha) - 2x)}{2L_1 + L(1 + \sin \alpha)} & \text{for } x \geq L_1 \sin \alpha + \frac{L \sin \alpha (1 + \sin \alpha)}{2} \\ \frac{\lambda g(L_1 + L \sin \alpha - x)(2L_1 + L(1 + \sin \alpha) + 2x)}{2L_1 + L(1 + \sin \alpha)} & \text{for } x \leq L_1 \sin \alpha + \frac{L \sin \alpha (1 + \sin \alpha)}{2} \end{cases}$$

A4

2.50 pts

Determine the displacement values $x_{max(1)}$ and $x_{max(2)}$ of the left end of the rope, at which it begins to lose contact with blocks 1 and 2, respectively, assuming that contact with the other block is maintained. Express your answers using L_1 , L and α .

Let us write down Newton's second law for a section of rope with mass $dm = \lambda dl$:

$$dm\vec{a} = d\vec{T} + dm\vec{g} + d\vec{N}.$$

For отрыве $d\vec{N} = 0$, therefore:

$$dm\vec{a} = d\vec{T} + dm\vec{g}.$$

Let ρ - radius кривизны consider surface, then $dl = \rho d\varphi$, where $d\varphi$ - rotation angle касательной to rope on consider section dl . Спроецируем equation of motion on direction нормали, direct to center кривизны:

$$dma_n = \lambda \rho d\varphi \cdot \frac{v^2}{\rho} = \lambda v^2 d\varphi = T d\varphi + \lambda \rho d\varphi \cdot g_n.$$

where g_n - projection acceleration free fall on direction нормали. Since radius block мал - condition отрыва following:

$$T = \lambda v^2.$$

Временно переобозначим quantity $x_{max(1)}$ and $x_{max(2)}$ over x_1 and x_2 correspondingly. Equation for x_1 following:

$$\frac{\lambda g(L_1 + x_1)(2L_1 + L(1 + \sin \alpha) - 2x_1)}{2L_1 + L(1 + \sin \alpha)} = \frac{2\lambda g x_1^2}{2L_1 + L(1 + \sin \alpha)}.$$

from which:

$$4x_1^2 - x_1 L(1 + \sin \alpha) - L_1(2L_1 + L(1 + \sin \alpha)) = 0.$$

Y This quadratic equation has one positive root, therefore:

The equation for x_2 is:

$$\frac{\lambda g(L_1 + L \sin \alpha - x_2)(2L_1 + L(1 + \sin \alpha) + 2x_2)}{2L_1 + L(1 + \sin \alpha)} = \frac{2\lambda g x_2^2}{2L_1 + L(1 + \sin \alpha)},$$

from which:

$$4x_2^2 + x_2 L(1 - \sin \alpha) - (L_1 + L \sin \alpha)(2L_1 + L(1 + \sin \alpha)) = 0.$$

This quadratic equation has one positive root, so:

Answer: Answer:

$$x_{max(1)} = \frac{L(1 + \sin \alpha)}{8} + \sqrt{\left(\frac{L(1 + \sin \alpha)}{8}\right)^2 + \frac{L_1(2L_1 + L(1 + \sin \alpha))}{4}}.$$

A5

1.20 pts

Calculate the speed of the rope at the moment of its separation from one of the blocks if the inclined plane forms angles $\alpha_1 = 30^\circ$, $\alpha_2 = 45^\circ$ and $\alpha_3 = 60^\circ$ with the horizon.

For each of the angles, separation is achieved at x , equal to the smaller of the values $x_{max(1)}$ and $x_{max(2)}$ corresponding to a given angle. Let's calculate them:

$\alpha, x_{max(1)}, m, x_{max(2)}, m$ 30 0.777 0.731 45 0.833 0.868 60 0.876 0.973

Thus, at angle α_1 the rope first loses contact with the second block, and at angles α_2 and α_3 - with the first. From here we find:

Answer: Answer:

$$v_1 = 2.068 \text{ m/s} \quad v_2 = 2.264 \text{ m/s} \quad v_3 = 2.314 \text{ m/s}.$$

B1

0.80 pts

Determine the displacement value x_0 of the left end of the rope corresponding to the equilibrium position of the system. Express your answer in terms of $m, \lambda, L_1, L, \alpha, g, k$ and F_0 .

Let's write down the equilibrium conditions for three sections of the rope:

$$T_1 = \lambda g(L_1 + x_0) + mg \quad T_2 - T_1 = \lambda g L \sin \alpha \quad T_2 = \lambda g(L_2 - x_0) + F_{\text{yup}},$$

from which:

$$F_{\text{yup}} = \lambda g(L \sin \alpha + L_1 + x_0 + x_0 - L_1 - L \sin \alpha) + mg = mg + 2\lambda g x_0.$$

Since $F_{\text{control}} = F_0 + kx_0$, we get:

Answer: Answer:

$$x_0 = \frac{mg - F_0}{k - 2\lambda g}.$$

B2**0.70 pts**

The equilibrium position from the previous paragraph exists if the spring stiffness exceeds some minimum value k_{min} . Identify k_{min} . Express your answer using $m, \lambda, L_1, L, \alpha, g, k$ and F_0 . Further, assume everywhere that $k > k_{min}$.

The denominator of the result obtained in point B1 must be positive, so we have the limitation:

$$k > 2\lambda g.$$

Quantity x_0 also ограничена length L_2 right vertical section rope, therefore:

$$x_0 = \frac{mg - F_0}{k - 2\lambda g} \leq L_1 + L \sin \alpha \Rightarrow k \geq 2\lambda g + \frac{mg - F_0}{L_1 + L \sin \alpha}.$$

The last limitation is more significant, so we have:

Answer: Answer:

$$k_{min} = 2\lambda g + \frac{mg - F_0}{L_1 + L \sin \alpha}.$$

B3**0.80 pts**

Find the change in the potential energy of the system ΔW_p when the left end of the rope is displaced by the amount x . Express your answer in terms of $m, \lambda, L_1, L, \alpha, g, k, F_0$ and x .

For quantities $\Delta W_{p(g)}$ and $\Delta W_{p(in)}$ we have:

$$\Delta W_{p(g)} = -mgx \quad \Delta W_{p(in)} = -\lambda g x^2.$$

Change potential energy deformation spring $\Delta W_{p(def)}$ equal area under graph $F_{control}(\Delta l)$ and constitute:

$$\Delta W_{p(def)} = F_0 x + \frac{kx^2}{2}.$$

Finally:

Answer: Answer:

$$\Delta W_p = \frac{(k - 2\lambda g)x^2}{2} - (mg - F_0)x.$$

B4**0.70 pts**

If we take the potential energy W_p at the equilibrium position equal to zero, then it can be represented in the form

$$W_p = \frac{A(x - B)^2}{2}.$$

Determine A and B. Answer express through $m, \lambda, L_1, L, \alpha, g, k$ and F_0 .

From the expression for ΔW_p we have:

$$W_p = \frac{(k - 2\lambda g)x^2}{2} - (mg - F_0)x + C = \frac{k - 2\lambda g}{2} \left(x - \frac{mg - F_0}{k - 2\lambda g} \right)^2 - \frac{(mg - F_0)^2}{2(k - 2\lambda g)} + C.$$

Determine potential energy in position equilibrium equal zero:

$$W_p = \frac{k - 2\lambda g}{2} \left(x - \frac{mg - F_0}{k - 2\lambda g} \right)^2.$$

Thus:

Answer: Answer:

$$A = k - 2\lambda g \quad B = \frac{mg - F_0}{k - 2\lambda g}.$$

B5

0.50 pts

Determine the maximum speed of the load v_{max} and its displacement from the initial position x_{max} if the system is released without an initial speed. Express your answers using $m, \lambda, L_1, L, \alpha, g, k$ and F_0 .

The maximum speed of the load is achieved at the point with coordinate x_0 and is determined from the law of conservation of mechanical energy:

$$\frac{(m + \lambda(2L_1 + L(1 + \sin \alpha))v_{max}^2}{2} = \frac{Ax_0^2}{2},$$

from where:

The value x_{max} is achieved at the moment the load stops, i.e. is determined from the condition:

$$\frac{A(x_{max} - x_0)^2}{2} = \frac{Ax_0^2}{2} \Rightarrow x_{max} = 2x_0,$$

from where:

Answer: Answer:

$$v_{max} = \frac{mg - F_0}{k - 2\lambda g} \sqrt{\frac{k - 2\lambda g}{m + \lambda(2L_1 + L(1 + \sin \alpha))}}.$$

B6

1.00 pts

What minimum speed v_{min} must the weight have in the initial position for the right end of the thread to reach the level of block 2? Express your answer using $m, \lambda, L_1, L, \alpha, g, k$ and F_0 .

The load must stop at the moment when it drops by the amount L_2 . From the law of conservation of mechanical energy:

$$\frac{A(L_2 - x_0)^2}{2} = \frac{Ax_0^2}{2} + \frac{(m + \lambda(2L_1 + L(1 + \sin \alpha))v_{min}^2}{2},$$

from:

Answer: Answer:

$$v_{min} = \sqrt{\frac{k - 2\lambda g}{m + \lambda(2L_1 + L(1 + \sin \alpha))}(L_1 + L \sin \alpha) \left(L_1 + L \sin \alpha - \frac{2(mg - F_0)}{k - 2\lambda g} \right)}$$